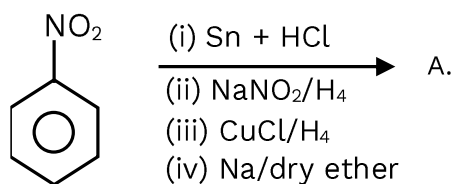


CHEMISTRY**Q.1** Compare boiling point of given solutions

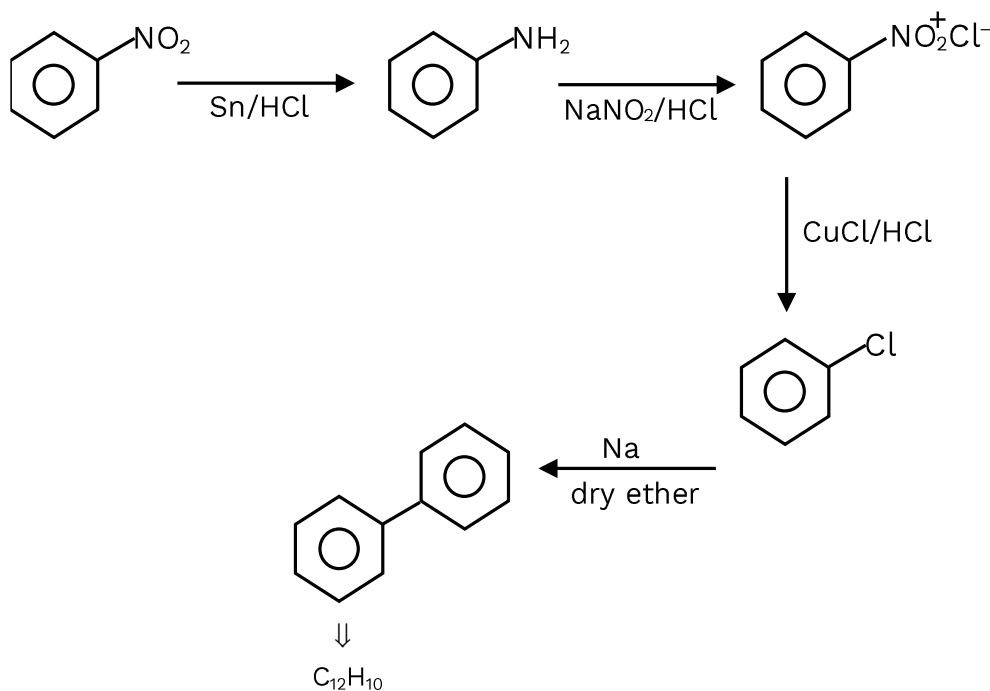
- (I) 10^{-4} NaCl (II) 10^{-3} NaCl (III) 10^{-2} NaCl (IV) 10^{-4} urea
 (1) I > II > III > IV (2) III > II > I > IV (3) II > I > III > IV (4) III > I > II > IV

Ans. (2)**Sol.** III > II > I > IV**Q2** Choose the incorrect order of electronegativity

- (1) S < Cl < O < F (2) Se < S < O < F (3) Al < Mg < Na (4) I < Br < Cl < F

Ans. (3)**Q3**

Find molecular weight of A

Ans. 154

$$12 \times 12 + 10 \Rightarrow 154$$

Q4. Number of linear compounds:
 I_3^- , NO_2 , O_3 , OF_2 , NO_2^+ , BeCl_2 , N_3^- , SO_3 , SO_2 , CO_2 , XeF_2
Ans. 6**Q5** The Ground state radius of Hydrogen atom is a_0 . Calculate the radius of first excited state of Helium ?

Sol. $R = \frac{R_0 n^2}{z}$ ground state $n = 1$

$$z = 1 \text{ for H}$$

$$R = R_0 \frac{1^2}{1}$$

$$R = R_0 = a_0$$

For He $\rightarrow$ excited $n = 2$

$$R = a_0 \frac{2^2}{2}$$

$$= 2a_0$$

Q6 Given the weight of the organic compound, 180 mg and the weight of the AgCl precipitated 143.5 gm. Calculate the estimation of Cl in percentage.

[Weight of Cl = 35.5 gm and Weight of Ag is 108 gm]

Sol. Correct the solution

180 mg of OC = 0.18g

143.5 mg = 0.1435 g

AgCl.

143.5 g AgCl $\longrightarrow$ 35.5 g Cl

$$0.1435\text{g} \longrightarrow \frac{35.5}{143.5} \times \frac{0.1435}{10000} \text{ g Cl}$$

$$= 35.5 \times 10^{-3} \text{ g Cl}$$

$$\% \text{ of Cl} \Rightarrow \frac{35.5 \times 10^{-3}}{0.18} \times 100 = 20\%$$

Q7 Which of the following lanthanide ion $4f^7$ electronic configuration

$\text{Eu}^{+3}, \text{Gd}^{+3}, \text{Eu}^{+2}, \text{Gd}^{+2}, \text{Sm}^{+2}, \text{Tb}^{+2}$

(1) $\text{Eu}^{+3}, \text{Gd}^{+2}$

(2) $\text{Gd}^{+3}, \text{Eu}^{+2}$

(3) $\text{Sm}^{+2}, \text{Tb}^{+2}$

(4) $\text{Gd}^{+2}, \text{Sm}^{+2}$

Ans. (2)

Q8 Which of the following can be used to prepare marshall's acid ($\text{H}_2\text{S}_2\text{O}_8$)

(1) Electrolysis of Conc. Na_2SO_4

(2) Electrolysis of dilute Na_2SO_4

(3) Electrolysis of Conc. H_2SO_4

(4) Electrolysis of dilute H_2SO_4

Ans. (4)

Sol. On electrolysis of dil H_2SO_4 product is H_2 , O_2 and $\text{H}_2\text{S}_2\text{O}_6$

$\underset{\text{(cathode)}}{1}$, $\underset{\text{(Anode)}}{2}$

Q9 **List-I**

(P) IE

(Q) Metallic character

(R) Electronegativity

(S) ionic radii

(1) $\text{P} \rightarrow 1; \text{Q} \rightarrow 4; \text{R} \rightarrow 2; \text{S} \rightarrow 3$

(3) $\text{P} \rightarrow 1; \text{Q} \rightarrow 2; \text{R} \rightarrow 4; \text{S} \rightarrow 3$

Ans. (1)

List-II

(1) $\text{B} < \text{C} < \text{O} < \text{N}$

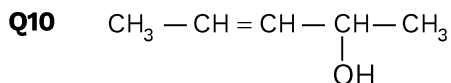
(2) $\text{Al} < \text{Si} < \text{P} < \text{S}$

(3) $\text{F}^- > \text{Na}^+ > \text{Mg}^{+2} > \text{Al}^{+3}$

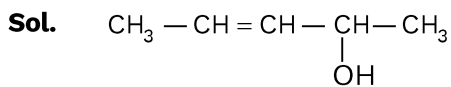
(4) $\text{K} > \text{Mg} > \text{Al} > \text{S}$

(2) $\text{P} \rightarrow 3; \text{Q} \rightarrow 4; \text{R} \rightarrow 2; \text{S} \rightarrow 1$

(4) $\text{P} \rightarrow 4; \text{Q} \rightarrow 1; \text{R} \rightarrow 2; \text{S} \rightarrow 3$

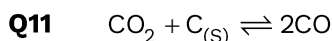
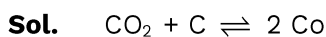
Sol.

Find out the total number of stereoisomers

Ans. (4)

$$n = 2$$

$$2^2 = (4)$$

Calculate Partial Pressure of $\text{CO}_{(\text{g})}$. If $\text{CO}_{2(\text{g})}$ at 0.5 atm is passed over heated carbon. Total pressure at equilibrium is 0.8.

$$0.5 \quad 0$$

$$0.5 - x$$

$$0_p = 0.5 - x + 2x$$

$$0.8 = 0.5 + 2x$$

$$x = 0.3 \text{ atm}$$

$$P_{\text{CO}_2} = 0.5 - 0.3 = 0.2$$

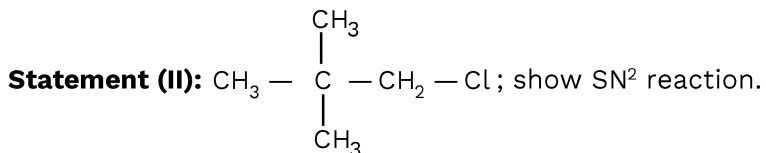
Q12 Which of the following is not true?

(1) decay constant does not depends on temperature.

(2) decay constant increases with temperature.

$$(3) t_{1/2} = \frac{\ln 2}{K}$$

(4) None

Ans. (2)**Q13** Given below are two statements :**Statement (I):** $\text{CH}_3\text{—O—CH}_2\text{—Cl}$; 1° alkyl halide show SN^1 reaction.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are incorrect

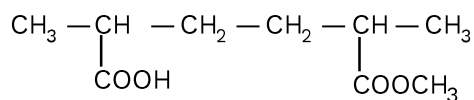
(2) Statement I is correct but Statement II is incorrect

(3) Both Statement I and Statement II are correct

(4) Statement I is incorrect but Statement II is correct

Ans. (1)

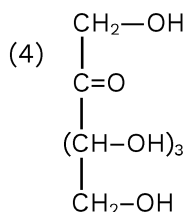
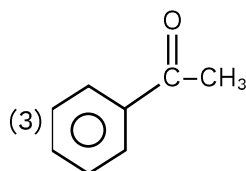
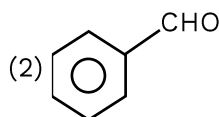
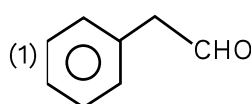
Q14 Correct IUPAC naming of the following



- (1) 5-methoxy Carbonyl-2-methyl hexanoic acid
- (2) 2-methoxy Carbonyl-5-methyl hexanoic acid
- (3) 2-methyl-5-methoxy Carbonyl hexanoic acid
- (4) 5-methoxy Carbonyl-1-methyl hexanoic acid

Ans. (1)

Q15 Which of the following gives positive test with Fehling's solution?



Ans. (1) and (4)

Q16 Ascorbic acid is which type of vitamin?

- (1) Vitamin A
- (2) Vitamin B
- (3) Vitamin C
- (4) Vitamin D

Ans. (1)

Q17 $[\text{Ni}(\text{CO})_4]$ and $[\text{NiCl}_4]^{2-}$ respectively have

- (1) $[\text{Ni}(\text{CO})_4]$ is square planar, diamagnetic while $[\text{NiCl}_4]^{2-}$ is Tetrahedral, Paramagnetic
- (2) $[\text{Ni}(\text{CO})_4]$ is tetrahedral, Paramagnetic while $[\text{NiCl}_4]^{2-}$ is Tetrahedral, diamagnetic
- (3) $[\text{Ni}(\text{CO})_4]$ is tetrahedral, diamagnetic while $[\text{NiCl}_4]^{2-}$ is Tetrahedral, Paramagnetic
- (4) $[\text{Ni}(\text{CO})_4]$ is tetrahedral, diamagnetic while $[\text{NiCl}_4]^{2-}$ is Square planar, diamagnetic

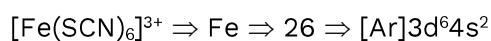
Ans. (3)

Q18 Which of the following compound have CFSE value is zero?

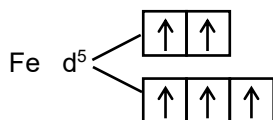
- (1) $\text{K}_4[\text{Fe}(\text{CN})_6]$
- (2) $\text{K}_3[\text{Fe}(\text{SCN})_6]$
- (3) $[\text{Fe}(\text{en})_3]\text{Cl}_3$
- (4) $[\text{Fe}(\text{NH}_3)_6]\text{Br}_2$

Ans. (2)

Sol. $\text{CFSE} = (-0.4 \times e^- t_{2g}) + (-0.6 \times e^- e_g) \Delta_0$



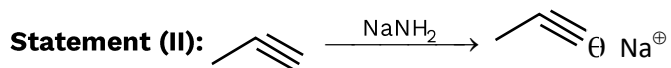
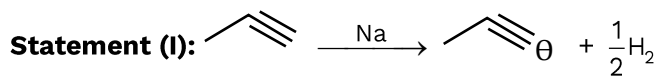
SCN is weak field lig and no pairing



$$\text{CFSE} = (-0.4 \times 3) + (0.6 \times 2)\Delta_0 = 0$$

(2) is correct.

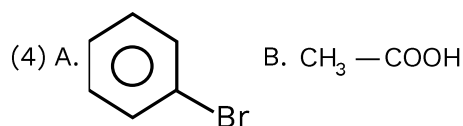
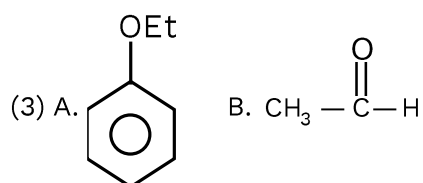
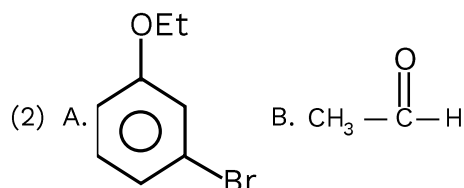
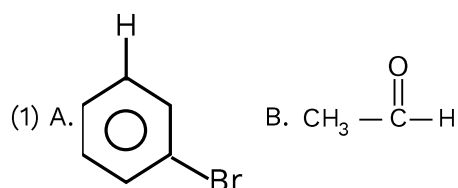
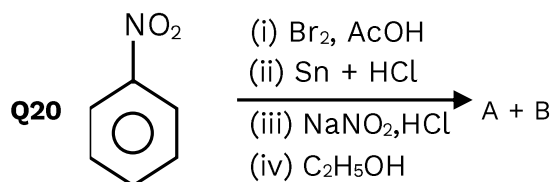
Q19 Given below are two statements :



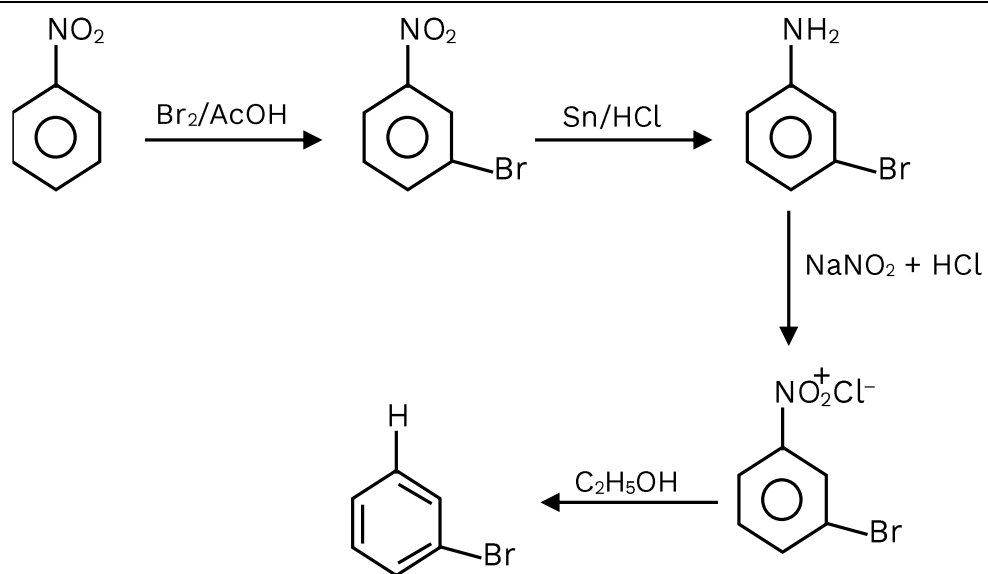
In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect
- (3) Both Statement I and Statement II are correct
- (4) Statement I is incorrect but Statement II is correct

Ans. (3)



Ans. (4)

Sol.

(4) is correct.